The Bug That Slept for Years — And the Fix That Brought Down Production

Uncovering a Silent Resource Leak with Elastic Observability

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Outline

1 The Setup

2 The Plot

3 The Investigation

4 Root Cause & Takeaways

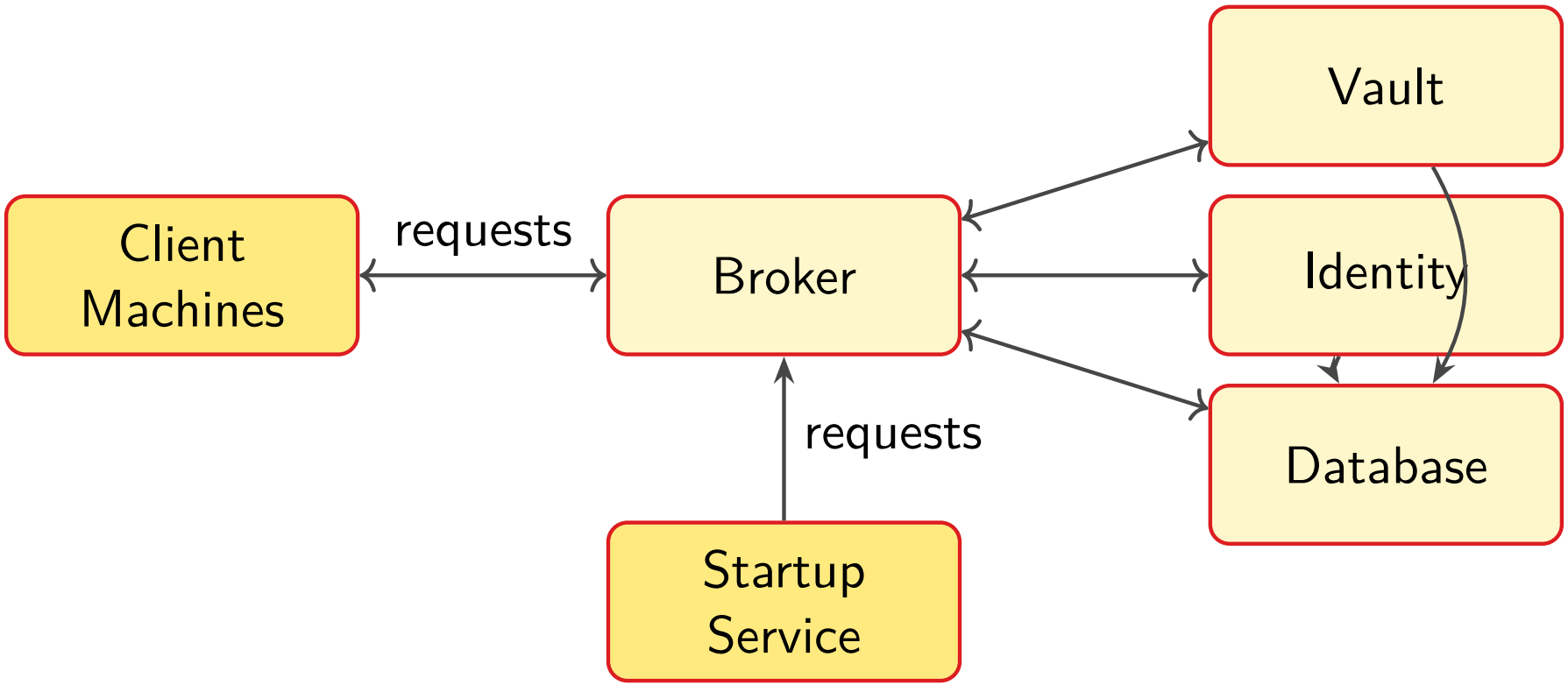
The Architecture & The Setup

System Landscape

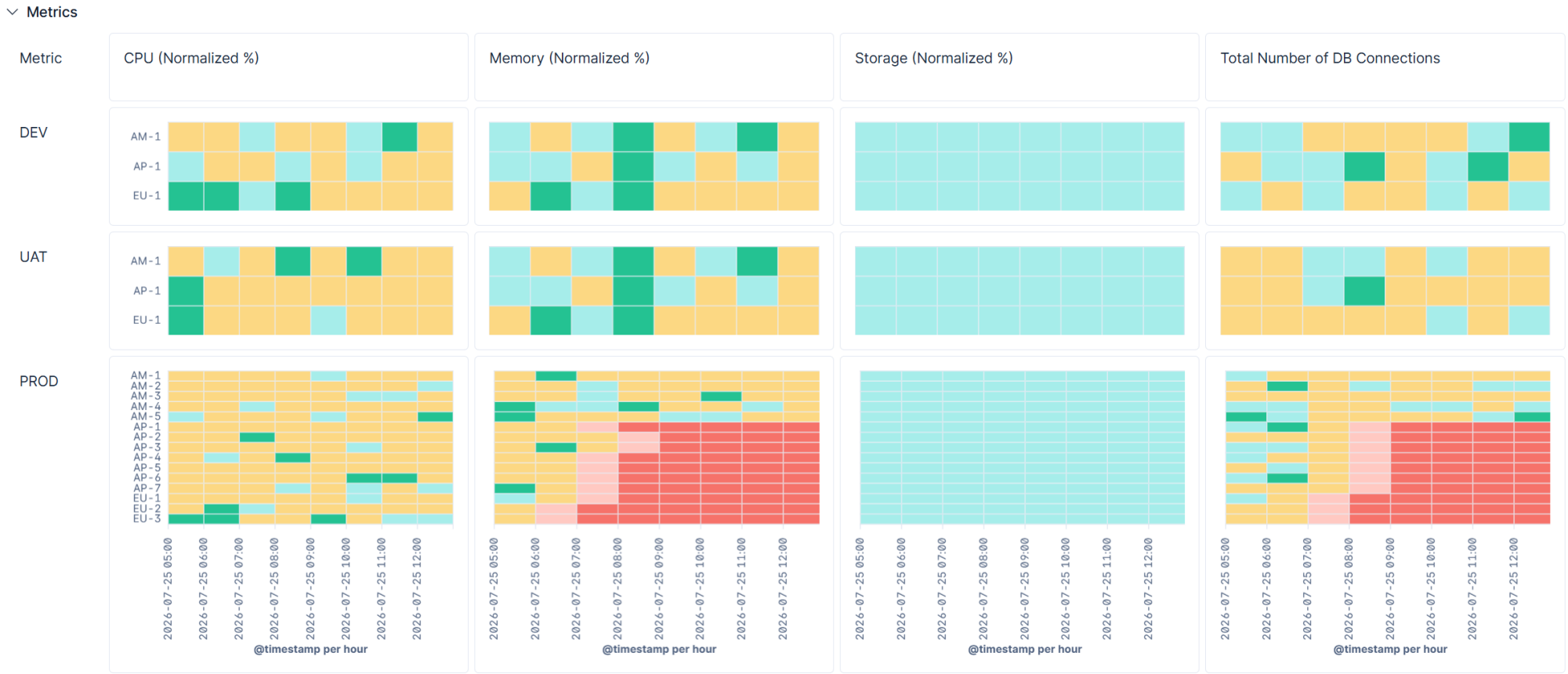
Complex multidiscipline backend ecosystem connecting multiple critical microservices.

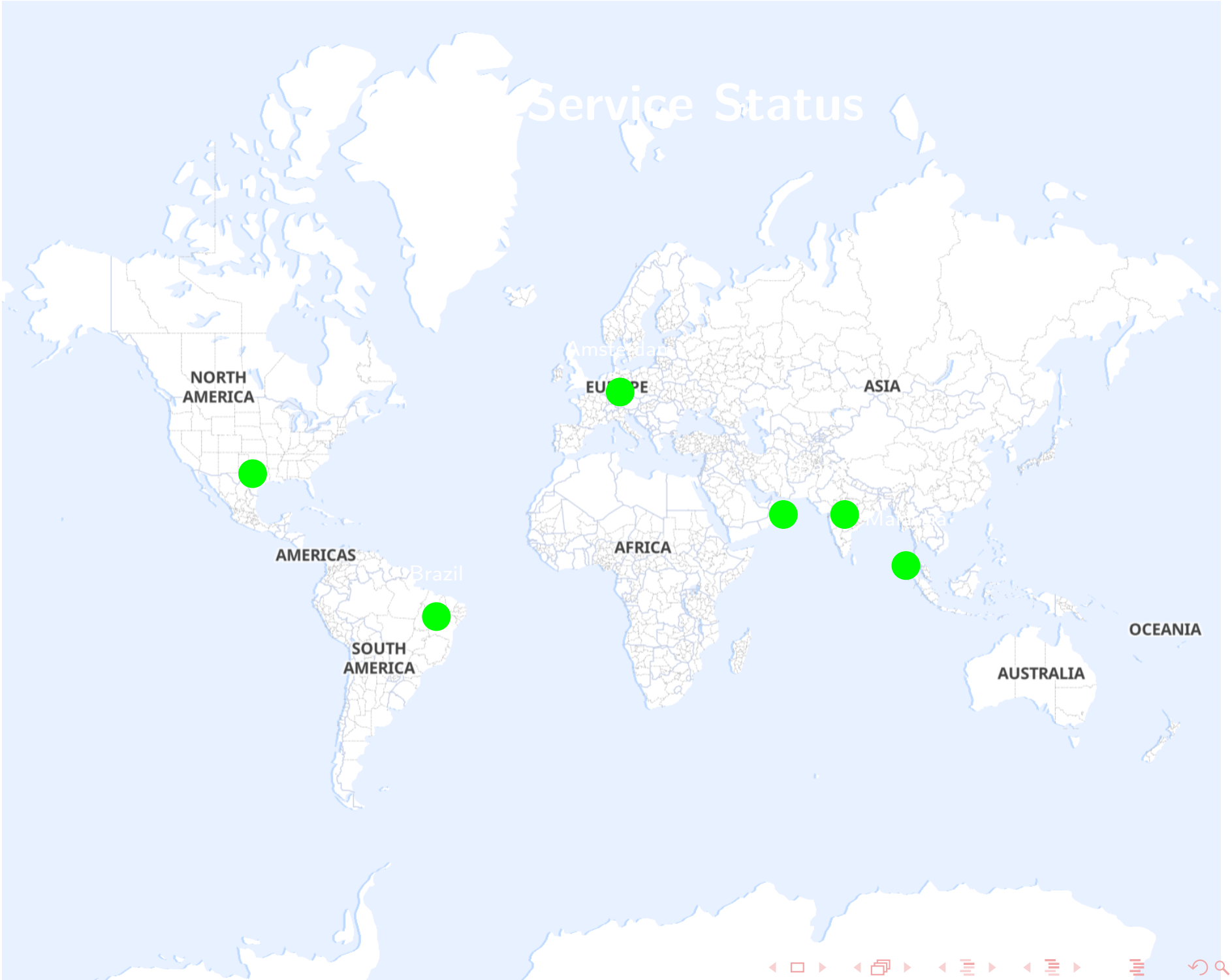
- **Startup:** Provisioning.
- **Vault:** Secure storage and management of secrets.
- **Identity:** Authentication and authorization.
- **Database:** Persistent data storage and management.
- **Broker:** Central communication hub between services.
- **Clients:** Backend consumers through the Broker.

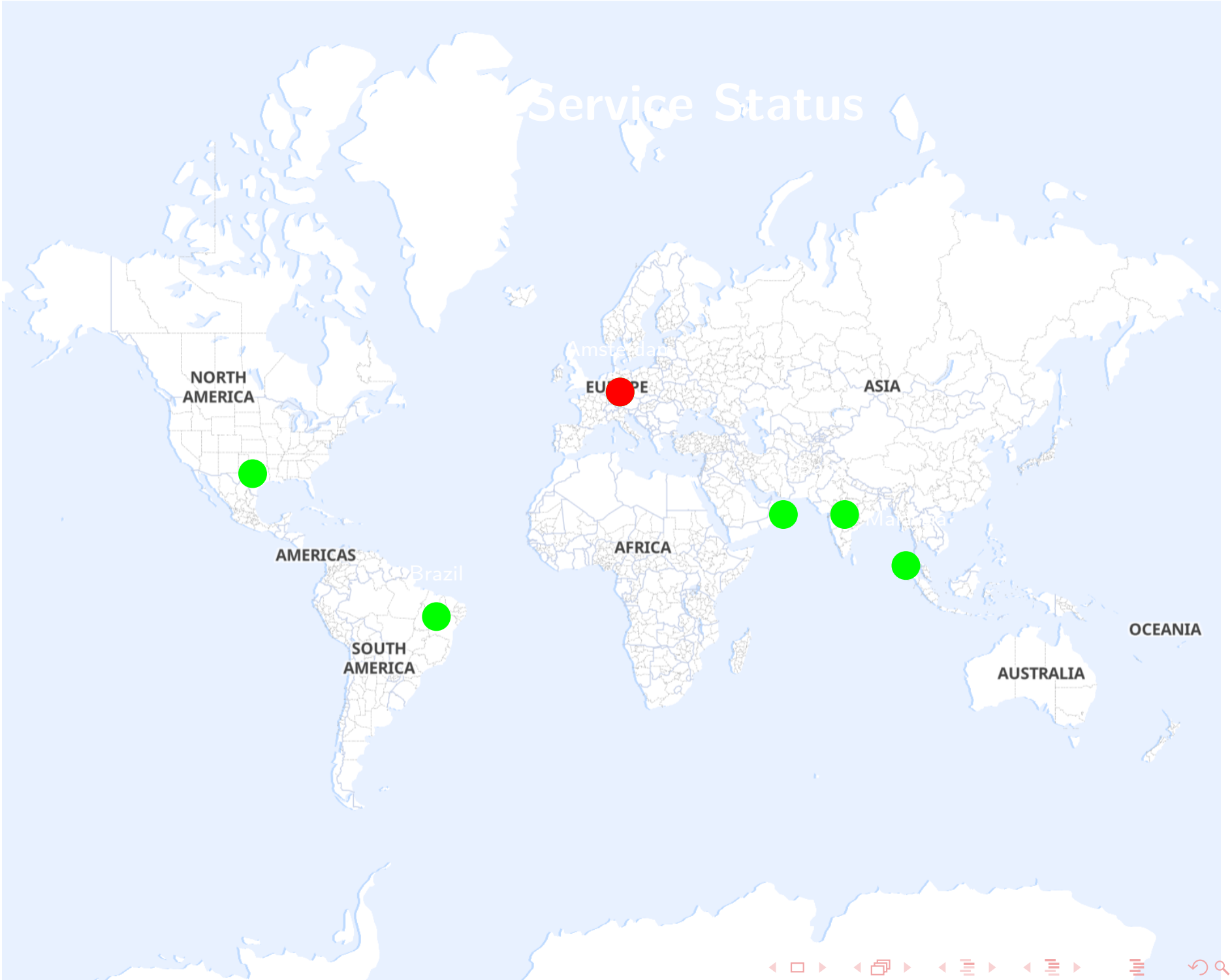
System Architecture Diagram

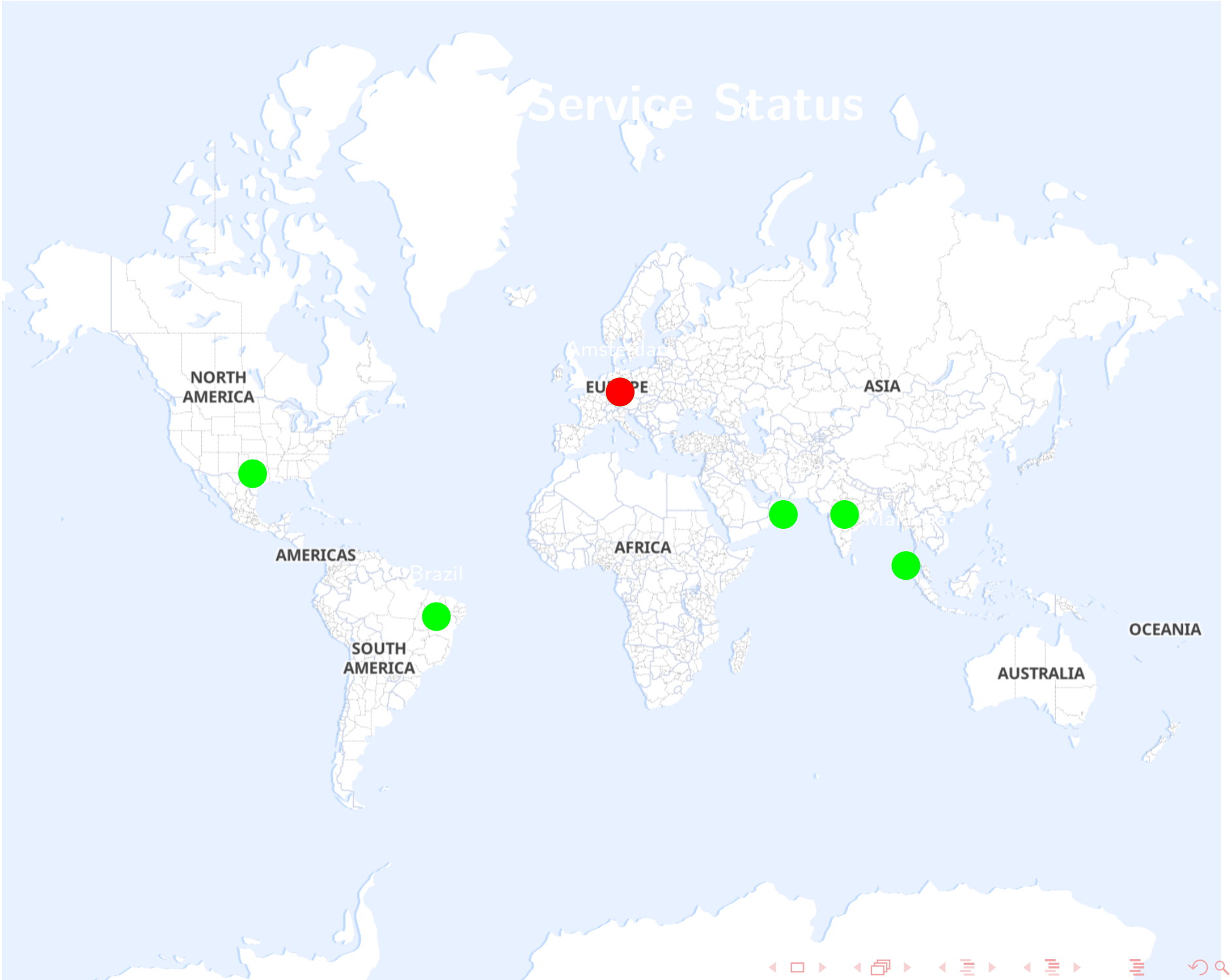


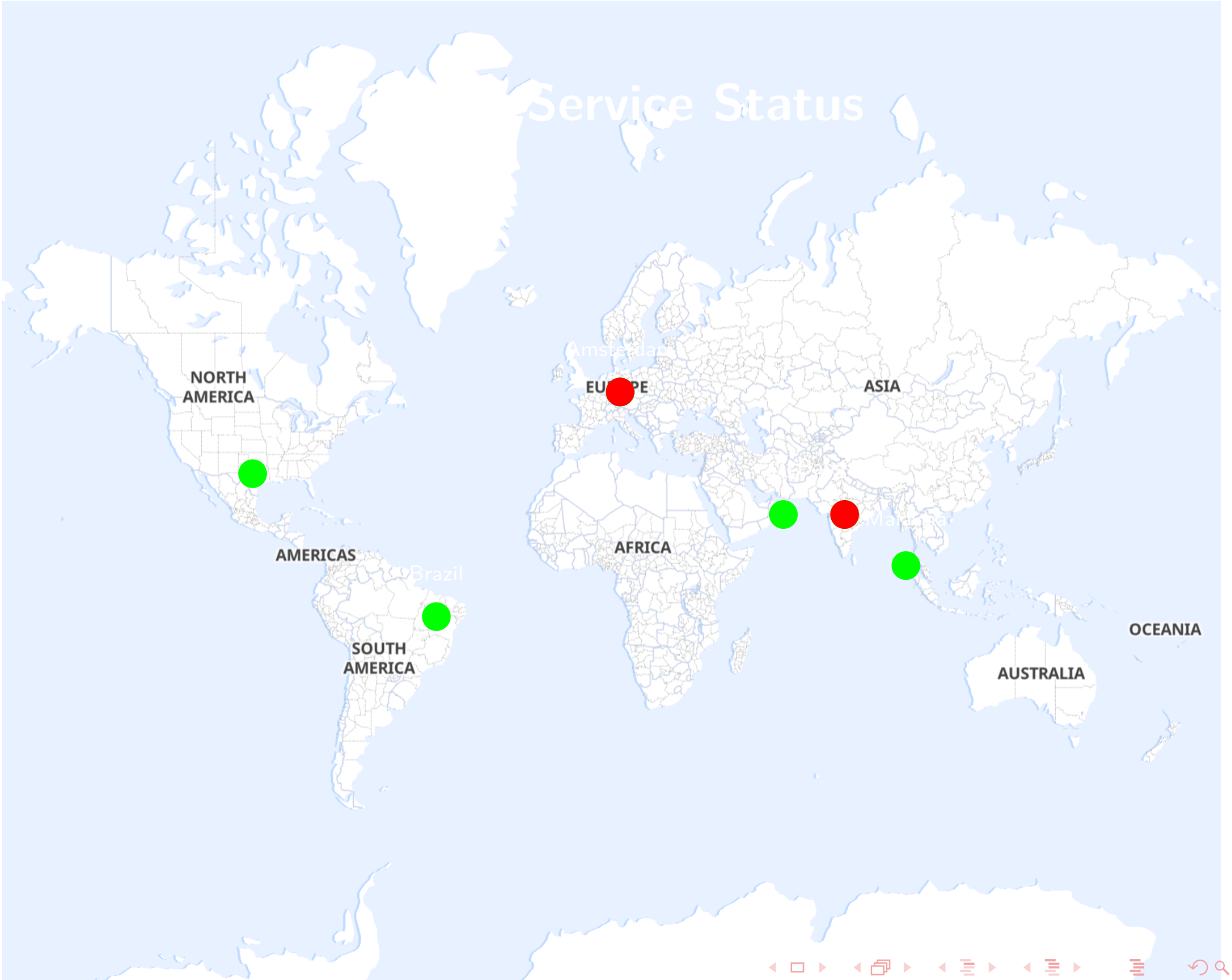
How it foldeded into

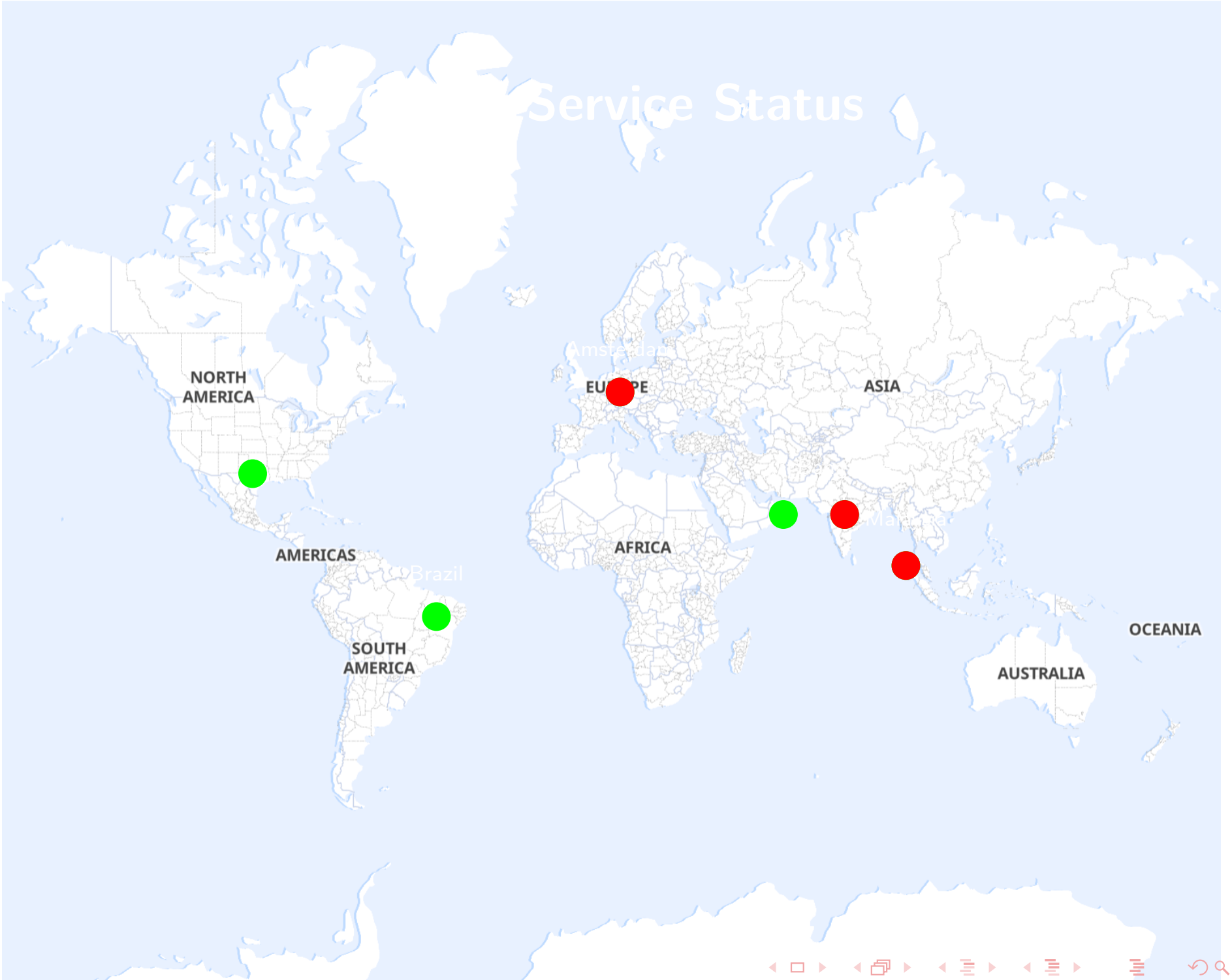


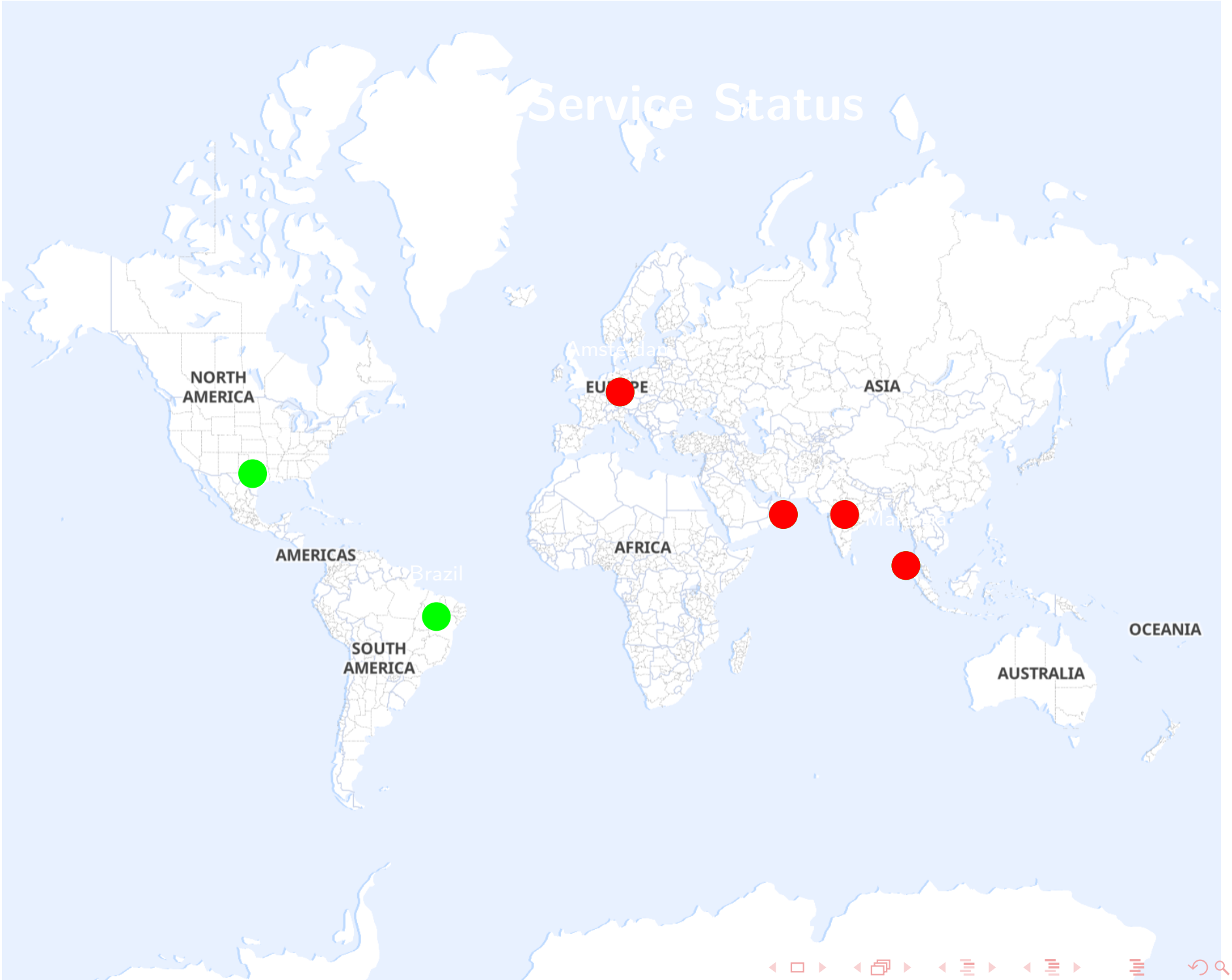

Service Status
NORTH AMERICA
AMERICAS
SOUTH AMERICA
Brazil
AFRICA
ASIA
OCEANIA
AUSTRALIA

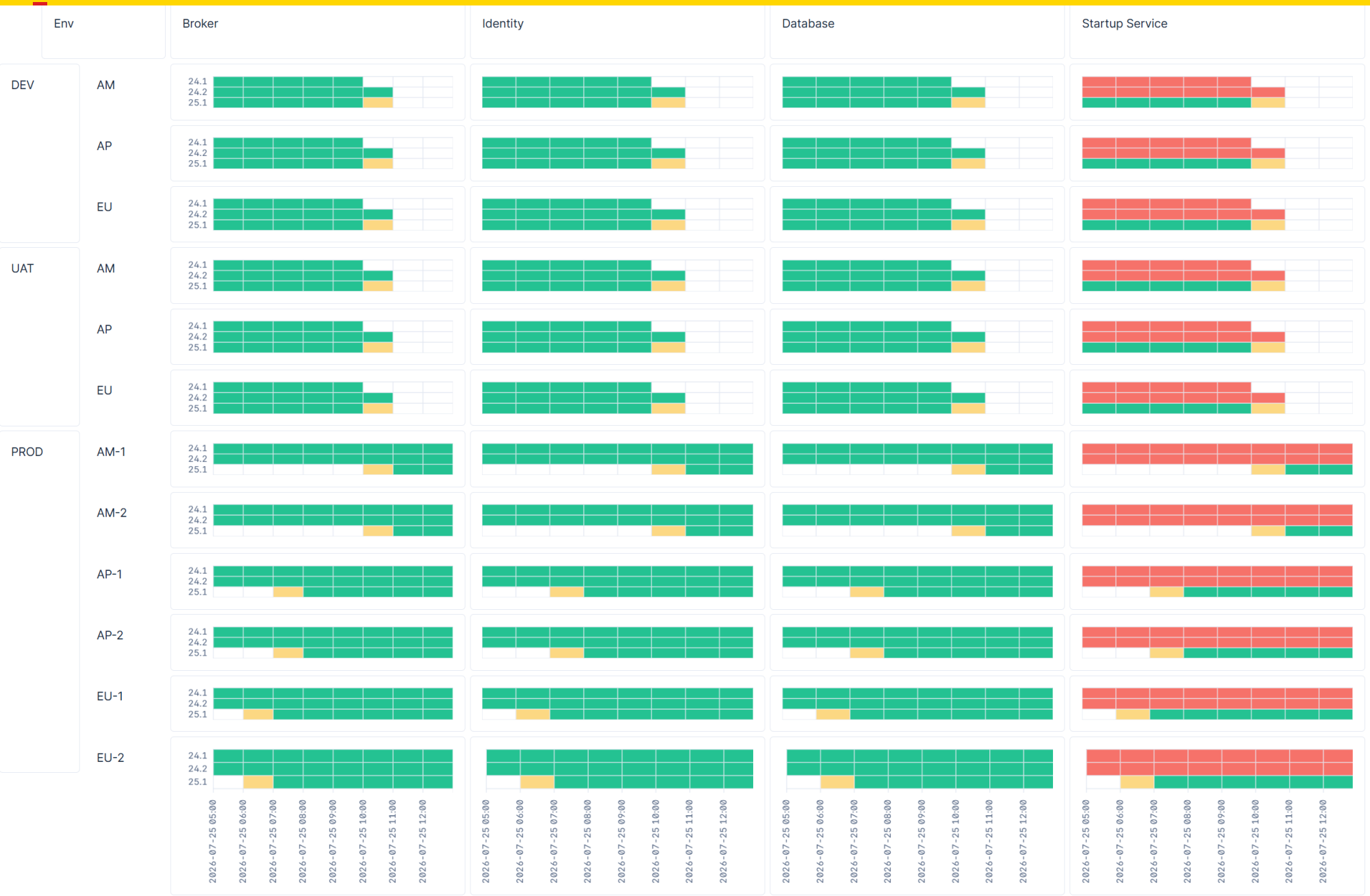

Service Status
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AUSTRALIA


Env
Broker
Identity
Database
Startup Service
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AM
AP
EU
UAT
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AP
EU
PROD
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AM-2
AP-1
AP-2
EU-1
EU-2
24.1
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2026-07-25 08:00
2026-07-25 09:00
2026-07-25 10:00
2026-07-25 11:00
2026-07-25 12:00

The Fog of War: False Leads

Initial triage was chaotic. We were blinded by three major red herrings:

- **The Geography Theory:** Europe and APAC collapsed.
- **The Infrastructure Theory:** Fresh troubles with our infrastructure provider.
- **The Environment Theory:** Why only production? DDoS, EDoS, or REA?

Eliminating the Leads

Using more detailed drilldown dashboards to provide better evidence gave us tools to eliminate possible candidates:

- **Bad Actor:** Single IP.
- **Network Failure:** Prior investigations showed no issues.
- **America's Collapse:** Discarded based on evidence.
- **System Collapse:** Hungry for resources.

The Only Lead Left

The only remaining lead was a single IP address per deployment.

- **Narrow Down to Agent Services:** Startup, Broker, Vault, and Identity.

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- **Create Our Own Beats:** Digging into the metrics and logs ourselves.

The Outcome

Following the results of the investigation, we determined:

- **Research Teams:** Inconclusive (indirectly pointing to startup service).
- **Results of the Beats:** Startup service.

All Eyes on the Startup Service

- **Red Herring:** Crying wolf.
- **Neglected:** Crash ignored.
- **Backlog:** No priority or manpower.
- **Missing Errors:** The elephant is missing in the room.

Back in Time

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- **Success:** Or was it?

Test the Theory: Temporary and Permanent Solutions

- **Stop the Service:** Resources freed.
- **Temporary Solution:** Kill the service after deployment.
- **Permanent Solution:** Singleton pattern.

The Architectural Mishap & Resolution

The Fix:

- Refactored the connection into a **Singleton pattern**.
- Introduced readily new metrics.

Key Lessons Learned:

- **Lifecycle Hygiene:** Proper resource management.
- **Beware the Bugfix:** Can expose hidden architectural flaws.
- **Observability is Essential:** You cannot fix what you cannot see—tracing connection lifecycles saved production.

A Final Reflection

Signal vs. Noise

When production breaks, we all have a tendency to look for the bugs we *expect* to see—like a regional network drop or a bad OS patch.

"When the system is burning and false leads are everywhere, how do you ensure your observability tools give you the signal you need to find the truth, rather than just more noise to get distracted by?"

Thank You

Questions?